

# Amrun port capacity: berths and stockyard size

AnyLogic discrete event model. Version 2 (Model8)

## 1. The problem

Amrun ships bauxite through a port that shuts for cyclones, swell and rain. The mine feeds a stockyard and ships load from it. When the port shuts and the mine keeps digging, the stockyard fills. Once full, the mine stops and those tonnes are lost. The mine can feed 150,000 tonnes a day. One berth cannot move that.

Question: how many berths and how much stockyard does the port need, and where does more stop helping?

## 2. What the model does

Ore moves in 5,000 t lots. Ships carry 80,000 t.

- Mine feeds 150,000 t/day to the stockyard
- A full stockyard stops the mine
- Ships queue at anchor for a berth
- Loading takes 21.3 hours per ship
- Weather stops the mine, port, or both
- Weather is modelled as a Markov chain

Ships arrive faster than the port can load. The berth is never idle, so the model returns a clean ceiling.

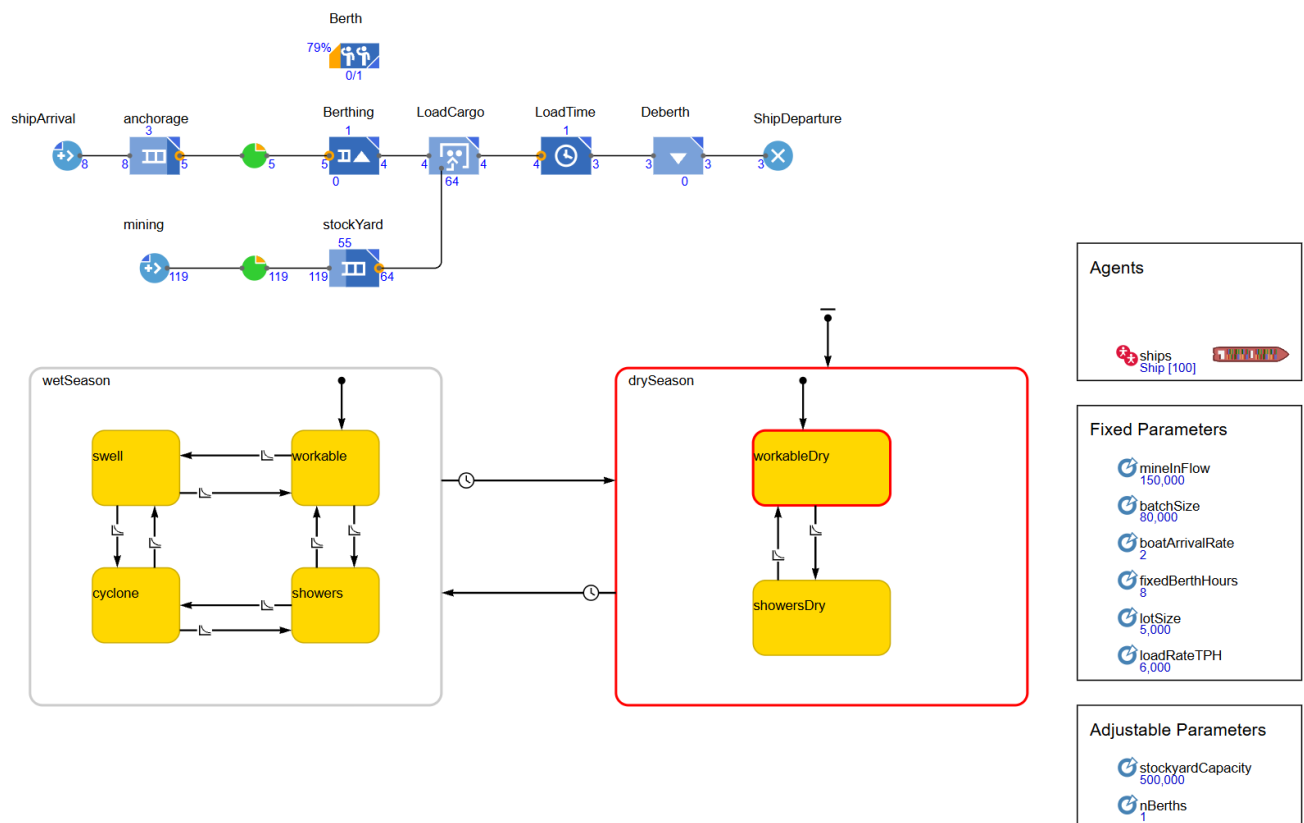


Figure 1. Model structure.

### 3. Approach: McKinsey seven steps

| Step            | What it meant here                              |
|-----------------|-------------------------------------------------|
| 1. Define       | How many berths and how much stockyard?         |
| 2. Disaggregate | Mine inflow, stockyard, berth outflow, weather. |
| 3. Prioritise   | Vary berths and stockyard only.                 |
| 4. Workplan     | Expect stockyard alone to fail. Sweep 36 runs.  |
| 5. Analyse      | Run the sweep. Check mass balance.              |
| 6. Synthesise   | Find where the curve flattens                   |
| 7. Communicate  | This report                                     |

### 4. What was run

36 runs. 12 stockyard sizes by 3 berth counts. 1,460 days each.

| Held fixed          | Value                                    |
|---------------------|------------------------------------------|
| Mine feed           | 150,000 t/day                            |
| Ship parcel         | 80,000 t                                 |
| Ship arrivals       | 2 per day                                |
| Loader rate         | 6,000 t/h                                |
| Fixed time at berth | 8 h                                      |
| Ore lot size        | 5,000 t                                  |
| Varied              | Levels                                   |
| Stockyard capacity  | 250,000 t to 3,000,000 t, step 250,000 t |
| Berths              | 1, 2, 3                                  |

Iterations completed: 36

#### Parameters

|                   |           |
|-------------------|-----------|
| mineInFlow        | 150,000   |
| batchSize         | 80,000    |
| boatArrivalRate   | 2         |
| stockyardCapacity | 3,000,000 |
| nBerths           | 3         |
| fixedBerthHours   | 8         |
| lotSize           | 5,000     |
| loadRateTPH       | 6,000     |

Figure 2. Sweep complete, all 36 runs.

## 5. Results

|    | A                                                                                                   | B              | C               | D               | E | F | G |
|----|-----------------------------------------------------------------------------------------------------|----------------|-----------------|-----------------|---|---|---|
| 1  | <b>Optimal Berth Count and Stockpile Size</b>                                                       |                |                 |                 |   |   |   |
| 2  | <i>Total exported over 4 years (Mt). Darker green = more throughput. boatArrivalRate = 2.0/day.</i> |                |                 |                 |   |   |   |
| 3  |                                                                                                     |                |                 |                 |   |   |   |
| 4  | <b>Stockpile (Mt)</b>                                                                               | <b>1 Berth</b> | <b>2 Berths</b> | <b>3 Berths</b> |   |   |   |
| 5  | 0.25                                                                                                | 125.6          | 195.8           | 195.9           |   |   |   |
| 6  | 0.50                                                                                                | 128.6          | 197.0           | 197.1           |   |   |   |
| 7  | 0.75                                                                                                | 129.2          | 197.7           | 197.7           |   |   |   |
| 8  | 1.00                                                                                                | 129.4          | 197.9           | 197.9           |   |   |   |
| 9  | 1.25                                                                                                | 129.8          | 198.2           | 198.2           |   |   |   |
| 10 | 1.50                                                                                                | 130.0          | 198.3           | 198.3           |   |   |   |
| 11 | 1.75                                                                                                | 130.0          | 198.3           | 198.3           |   |   |   |
| 12 | 2.00                                                                                                | 130.0          | 198.3           | 198.3           |   |   |   |
| 13 | 2.25                                                                                                | 130.0          | 198.3           | 198.3           |   |   |   |
| 14 | 2.50                                                                                                | 130.0          | 198.3           | 198.3           |   |   |   |
| 15 | 2.75                                                                                                | 130.0          | 198.3           | 198.3           |   |   |   |
| 16 | 3.00                                                                                                | 130.0          | 198.3           | 198.3           |   |   |   |

|                             |                                         |
|-----------------------------|-----------------------------------------|
| Throughput at optimum       | 198.32 Mt over 4 years                  |
| Gain from 1st → 2nd berth   | +68.3 Mt (+52.6%)                       |
| Gain from 2nd → 3rd berth   | +0.0 Mt                                 |
| Gain from 1.5 → 3.0 Mt pile | +0.0 Mt                                 |
| Why it stops here           | Mine is maxed, mineAchievedFrac = 0.906 |

◀ **OPTIMUM: 2 berths, 1.5 Mt stockpile**

Figure 3. Tonnes exported over four years.

## 6. Results

1. The berth is the constraint, not the stockyard. One berth cannot load what the mine digs, so the pile sits full and the mine keeps stopping. A bigger pile only delays that.
2. Two berths are enough. A third stands idle waiting for ore.
3. The stockyard stops paying near 1.4 Mt. Above that the pile never fills and stack-out disappears.
4. What is left is the mine, not the port. Wet weather stops the mine directly, and port capacity cannot recover those tonnes.

## 7. Checks

- Mass balance holds in 31 of 36 runs
- Residual is 0.03% of tonnes mined
- One berth ceiling matches hand calculation
- Mine ceiling matches weather downtime

## 8. Moving forward

Further Analysis, build two berths and 1.5 Mt of stockyard. Skip the third berth.

- Repeat each point with ten replications
- Vary the loader rate, currently untested
- Add cost.

---

Not covered: blending, breakdowns, tides, ship diversion, capital cost, demurrage.